



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. G. Mamatha
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Roll Number	2520030054
Branch	CSE
Course Outcome	CO2 B+Trees,B-Trees,Fenwick Trees,Segment Trees

Case Study Topic :Airline Reservation Database Indexing and Passenger Analytics using B+ Trees and Fenwick Trees

Work Done Summary

In this case study, we implemented B-Tree, B+ Tree, Segment Tree, and Fenwick Tree concepts for airline reservation and passenger analytics systems.

B-Tree and B+ Tree were used for flight reservation database indexing and efficient record retrieval. Fenwick Tree was used to calculate passenger booking totals quickly, while Segment Tree was analyzed for range-based flight occupancy queries.

The structures were compared based on search efficiency, storage optimization, and query performance. The study concluded that B+ Trees are best suited for airline database indexing, while Fenwick Trees are effective for passenger count analytics and cumulative booking queries.

Implementation Details

B-Tree

1. Inserted flight reservation records into a multi-way search tree.
2. Allowed multiple keys per node.
3. Reduced tree height compared to BST.
4. Supported efficient disk-based indexing.

B+ Tree

1. Stored reservation records in leaf nodes.
2. Connected leaf nodes using linked traversal.
3. Enabled efficient range searches.
4. Used for airline reservation database indexing.

Fenwick Tree

1. Maintained passenger booking counts.
2. Supported prefix sum queries.
3. Updated booking counts dynamically.
4. Reduced query complexity.

Segment Tree

1. Performed range occupancy analysis.
2. Supported interval queries.
3. Handled flight seat utilization calculations.
4. Compared against Fenwick Tree performance.

Result :Screenshots / Output

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AIRLINE RESERVATION DATABASE INDEXING SYSTEM
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Reservation IDs:
1001, 1005, 1010, 1015, 1020, 1025, 1030

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B+ TREE CONSTRUCTION
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      [1010 | 1020]
      /   |   \
[1001 1005] [1010 1015] [1020 1025 1030]

Leaf Node Links:

[1001 1005] -> [1010 1015] -> [1020 1025 1030]

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SEARCH OPERATION
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Searching Reservation ID = 1025

Result:
Reservation Found
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RANGE QUERY
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Reservations between 1010 and 1030

1010
1015
1020
1025
1030

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FENWICK TREE ANALYSIS
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Passenger Counts:
120 150 180 200 170 210 190 160

Total Passengers (FL101 - FL105)
820

=====
Process Finished Successfully
=====
```

Time Complexity Analysis

Operation	B-Tree	B+ Tree	Segment Tree	Fenwick Tree
Search	$O(\log n)$	$O(\log n)$	-	-
Insert	$O(\log n)$	$O(\log n)$	-	$O(\log n)$
Range Query	$O(\log n + k)$	$O(\log n + k)$	$O(\log n)$	$O(\log n)$
Update	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$

Conclusion

The study demonstrated the application of B-Trees, B+ Trees, Segment Trees, and Fenwick Trees in airline reservation and analytics systems. B+ Trees proved most suitable for airline database indexing because of efficient range retrieval and disk access optimization. Fenwick Trees provided fast passenger count aggregation and update operations. These structures are widely used in airline reservation systems, airport databases, and flight analytics platforms.

GITHUB LINK: https://github.com/manishaperugu49-glitch/2520030054-DSA_2_CaseStudy_CO2

Student Signature	Faculty Signature